

Construction and testing

Download Arduino IDE from www.arduino.cc and complete the installation process. Open Arduino IDE and install the libraries. Download the source code file and compile Sketch using Arduino IDE. Connect Arduino Uno to your computer and select the proper port and board. Upload Sketch (ADCDAC_Sketch.ino). For further information, visit www.arduino.cc

Assemble all the components on a general-purpose PCB with a proper soldering iron. An actual-size, single-side PCB layout for the trainer circuit is shown in Fig. 2 and its component layout in Fig. 3. Make all the wiring as per the circuit diagram (see Fig. 1). Cross-check for any wrong connections. Remove the programmed ATmega 328P IC from Arduino and insert in its socket (ensure proper orientation). (**Note.** I2C module and 20x4 LCD need to be connected externally using jumper wire as their footprints are not given in PCB layout.)

When you switch on 230V power supply through switch S9, the LCD will show the welcome messages written in the software. The LCD will thereafter show all the further procedures.

To calibrate, switch on the unit and ensure switch S9 is in ADC mode, that is, it is connected to the 5V power supply. Connect a multimeter across test points TP3 and TP4 and select 20V DC range.

Rotate pot VR1 fully anticlockwise; all the LEDs should now be off. If they are not off, adjust pot VR2 till all of them are off. The multimeter reading should be 0V in this position.

Next, rotate pot VR1 fully clockwise and ensure all the eight LEDs

PARTS LIST

Semiconductors:

IC1	- 7805, 5V voltage regulator
IC2	- ATmega 328P (with Arduino Boot loader)
D1-D8	- 1N4148 signal diode
D9-D12	- 1N4007 rectifier diode
ZD1	- 1N4733 diode
LED1-LED9	- 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1	- 10-kilo-ohm
R2-R12	- 1-kilo-ohm
R1-R9	- 5.6-kilo-ohm
R10	- 10-kilo-ohm
R11	- 22-kilo-ohm
VR1	- 10-kilo-ohm 10-turn pot
VR2	- 2-kilo-ohm pot

Capacitors:

C1, C4, C9, C10	- 10nF ceramic disc
C2, C3, C5	- 100uF, 25V electrolytic
C6, C7	- 22pF ceramic disc
C8	- 10uF, 25V electrolytic

Miscellaneous:

S1-S9	- SPST mini toggle switch
S10(a) and S10(b)	- DPST mini toggle switch
J1	- Terminal block (2-way)
XT1	- 16MHz crystal
F1	- 500mA fuse
X1	- 230V primary to 9V-0-9V, 500mA transformer
MOD1	- I2C module
LCD1	- 20x4 LCD

(LED1-LED8) are lit up. The multimeter should show 5V.

Shift switch S10 to DAC mode by opening it. Connect the multimeter across test points TP1 and TP2. Keep all switches S1 through S8 at off (low) position. The eight LEDs (LED1 through LED8) should be in off condition and the multimeter should show 0V.

Shift the seven switches S1 through S7 to off (high) position. The seven LEDs (LED1 through LED7) will light up and multimeter will show 2.5V. Switch on all the eight switches S1 through S8 and all the eight LEDs (LED1 through LED8) will now light up and multimeter will show 5V. **EFY**

A. Asokan Ambali (left) is Chargeman (Instrument) in ECE Faculty, Indian Naval Academy, Ezhimala, Kannur. And Chinchu Grace Lukose (right), M.Tech is an Asst. Professor (ECE) in ECE Faculty, Indian Naval Academy, Ezhimala, Kannur



Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

Amaze Yourself!

Read Electronics For You

You'll realise how much you didn't know, and didn't know that you didn't know!

To Subscribe:
<http://subscribe.efy.in>

Scan This Code



EFY Note

The source code of this project is available for free download at www.electronicsforu.com